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Instruction: Answer SIX questions in the provided space. Justify your answer in each question, and in order to get a full mark, you must demonstrate all necessary steps.

Question 1 (6 marks)

a) Let f be a one-to-one function whose inverse function is given by the formula

$$f^{-1}(x) = x^5 + 2x^3 + 3x + 1, \text{ find } f(7). \quad [2]$$

Solution:

$$\text{let } y = f^{-1}(x) = x^5 + 2x^3 + 3x + 1$$

we swap x and y :

$$x = y^5 + 2y^3 + 3y + 1 \Leftrightarrow y = f(x)$$

$$\text{so } 7 = y^5 + 2y^3 + 3y + 1$$

$$y^5 + 2y^3 + 3y - 6 = 0$$

$$\Rightarrow y = 1$$

Since f is a one-to-one function,
Then we can get $f(7) = 1$.

b) Find the value k for which the limit $\lim_{x \rightarrow 3} \frac{4x^2 + kx + 7k - 6}{2x^2 - 5x - 3}$ exists. Clearly justify your answer. [4]

Solution:

$$2x^2 - 5x - 3 = (2x + 1)(x - 3)$$

Since the limit to exist as $x \rightarrow 3$

$\Rightarrow 4x^2 + kx + 7k - 6$ must be 0 at $x = 3$

$$\Rightarrow 4(3)^2 + 3k + 7k - 6 = 0$$

$$30 + 10k = 0$$

$$k = -3$$

$$\begin{aligned} \Rightarrow \lim_{x \rightarrow 3} \frac{4x^2 - 3x - 27}{2x^2 - 5x - 3} &= \lim_{x \rightarrow 3} \frac{(4x + 9)(x - 3)}{(2x + 1)(x - 3)} \\ &= \lim_{x \rightarrow 3} \frac{4x + 9}{2x + 1} \\ &= 3 \end{aligned}$$

Question 2 (4 marks)

Determine the following limit, if exists. (*L'Hopital's Rule is NOT allowed on this problem.*)

a) $\lim_{x \rightarrow 0^-} \tan^{-1}\left(\frac{1}{x}\right)$ [2]

b) $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\sin \theta}$ [2]

Solution:

$$\begin{aligned} \text{(a)} \quad x \rightarrow 0^- &\Rightarrow \frac{1}{x} \rightarrow -\infty \\ &\Rightarrow \tan^{-1}\left(\frac{1}{x}\right) \rightarrow -\frac{\pi}{2} \\ \text{Therefore } \lim_{x \rightarrow 0^-} \tan^{-1}\left(\frac{1}{x}\right) &= -\frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\sin \theta} &= \lim_{\theta \rightarrow 0} \frac{\frac{\sin \theta}{\cos \theta}}{\sin \theta} \\ &= \lim_{\theta \rightarrow 0} \frac{1}{\cos \theta} \\ &= 1 \end{aligned}$$

Question 3 (5 marks)

Find the derivative of the following.

a) $f(x) = \sin(x) + x^2 \tan^{-1}(x)$ [2]

b) $g(x) = (2x)^{2x}$ [3]

Solution:

$$(a) \quad (x^2)' = 2x, \quad (\tan^{-1}x)' = \frac{1}{1+x^2}$$

$$f'(x) = \cos x + 2x \tan^{-1}x + \frac{x^2}{1+x^2}$$

$$(b) \quad \ln g(x) = 2x \ln(2x)$$

$$\frac{g'(x)}{g(x)} = 2 \ln(2x) + 2x \cdot \frac{2}{2x} = 2 \ln(2x) + 2$$

$$g'(x) = (2x)^{2x} [2 \ln(2x) + 2]$$

Question 4 (10 marks)

a) Calculate $y'(x)$ for $\cos(x^2 + 2y) + xe^{y^2} = x$.

[4]

Solution:

$$\begin{aligned} -\sin(x^2 + 2y) \cdot (2x + 2y') + e^{y^2} + x e^{y^2} \cdot 2y \cdot y' &= 1 \\ -2x \sin(x^2 + 2y) - 2 \sin(x^2 + 2y) \cdot y' + e^{y^2} + 2xy e^{y^2} \cdot y' &= 1 \\ [2xy e^{y^2} - 2 \sin(x^2 + 2y)] y' &= 1 + 2x \sin(x^2 + 2y) - e^{y^2} \\ y' &= \frac{1 + 2x \sin(x^2 + 2y) - e^{y^2}}{-2 \sin(x^2 + 2y) + 2xy e^{y^2}} \end{aligned}$$

b) Use logarithmic differentiation to find the derivative of the function,

$$y = \frac{(\sqrt{5x+3})(\sqrt[3]{1-9\cos(4x)})}{\sqrt[4]{x^2-10x}}.$$

[6]

Solution:

$$\begin{aligned} \ln y &= \frac{1}{2} \ln(5x+3) + \frac{1}{3} \ln(1-9\cos 4x) - \frac{1}{4} \ln(x^2-10x) \\ \frac{y'}{y} &= \frac{1}{2} \cdot \frac{5}{5x+3} + \frac{1}{3} \cdot \frac{36 \sin 4x}{1-9\cos 4x} - \frac{1}{4} \cdot \frac{2x-10}{x^2-10x} \\ \frac{y'}{y} &= \frac{5}{2(5x+3)} + \frac{12 \sin 4x}{1-9\cos 4x} - \frac{x-5}{2(x^2-10x)} \\ y' &= \frac{\sqrt{5x+3}(\sqrt[3]{1-9\cos 4x})}{\sqrt[4]{x^2-10x}} \cdot \left[\frac{5}{2(5x+3)} + \frac{12 \sin 4x}{1-9\cos 4x} - \frac{x-5}{2(x^2-10x)} \right] \end{aligned}$$

Question 5 (5 marks)

Use L'Hospital's Rule to evaluate $\lim_{x \rightarrow \infty} (e^x + x)^{\frac{1}{x}}$.

[5]

Solution:

$$\text{let } y = (e^x + x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln(e^x + x)$$

Apply L'Hospital's Rule:

$$\lim_{x \rightarrow \infty} \frac{\ln(e^x + x)}{x} = \lim_{x \rightarrow \infty} \frac{e^x + 1}{e^x + x} = 1$$

$$\text{Therefore } \lim_{x \rightarrow \infty} y = e^1 = e$$

Question 6 (10 marks)

Given $f(x) = x^{\frac{4}{3}}(x-2)$

a) Identify the intervals on which f is increasing or decreasing. [8]

b) Identify the local maximum and minimum points of f , if any. [2]

Solution:

$$\begin{aligned} (a) \quad f'(x) &= \frac{4}{3} x^{\frac{1}{3}}(x-2) + x^{\frac{4}{3}} \\ &= x^{\frac{1}{3}} \left[\frac{4}{3}(x-2) + x \right] \\ &= \frac{1}{3} x^{\frac{1}{3}} (7x-8) \end{aligned}$$

Set $f'(x) = 0$ we can get:
 $x = 0$ or $x = \frac{8}{7}$

① $x < 0$:

$$x^{\frac{1}{3}} < 0, 7x-8 < 0 \Rightarrow f'(x) > 0 \Rightarrow \text{increasing}$$

② $0 < x < \frac{8}{7}$:

$$x^{\frac{1}{3}} > 0, 7x-8 < 0 \Rightarrow f'(x) < 0 \Rightarrow \text{decreasing}$$

③ $x > \frac{8}{7}$:

$$x^{\frac{1}{3}} > 0, 7x-8 > 0 \Rightarrow f'(x) > 0 \Rightarrow \text{increasing}$$

Therefore f is increasing on $(-\infty, 0) \cup (\frac{8}{7}, +\infty)$
 and is decreasing on $(0, \frac{8}{7})$.

(b) from (a) we can know

$$f \text{ has local maximum } f(0) = 0^{\frac{4}{3}}(-2) = 0$$

$$f \text{ has local minimum } f\left(\frac{8}{7}\right) = \left(\frac{8}{7}\right)^{\frac{4}{3}}\left(\frac{8}{7}-2\right) = -\frac{96}{49\sqrt[3]{7}}$$

Therefore local maximum point $\Rightarrow (0, 0)$

local minimum point $\Rightarrow \left(\frac{8}{7}, -\frac{96}{49\sqrt[3]{7}}\right)$.

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